

DIGGING FOR KNOWLEDGE

Final Report & Prototype Instruction Manual

Team member:

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Figure 1. An image of our product

Executive Summary:

The goal of the project was to have an interactive tool that aids in teaching STEM-related concepts to grades 4-6. The design was required to be reusable, reliable, and interactive, which was achieved through the use of servo motors, a keypad, and an Arduino to challenge the students to correctly input a numerical answer corresponding to its box. The box would contain parts of a whole pyramid for the students to construct, adding a layer of interactivity to the design. The project was also constrained to be challenging yet easy to store with the constraint of a \$200 budget, which was achieved by utilizing a plastic container to hold all its material and filled with shredded papers (Figure 1). In order to challenge the students, Ms. Weaver, the teacher for this project, will supply question cards that correspond to the boxes so that they can be opened using the keypad.

User's Guide

Assembly Instruction:

1. Place all of the puzzle pieces inside the boxes, leave the triangular base somewhere inside the box (let the students find it).
2. Plug 9 volt battery into the battery plug on the side of the box.
3. Create question cards for each of the boxes and input their answers into the keypad. Be sure to test each of the answers, it is always possible that the keypad did not register your input.
4. Scramble the boxes, be sure not to stretch them.

Operations:

To select a box, locate the indicated letter on the box of choice then use A, B, C, D, or A followed by B on the keypad.

After selecting a box, input a code of any digit then press # to confirm code input. At the initial start, all boxes are set with the default code 1234.

To close and clear the box inputs, press *.

To change the code of the selected servo enter the code 0000# to enter operation mode. In operation mode, input new code then select # to confirm new code.

Remember, the codes DO NOT have to be four digits long.

Troubleshooting Issues:

The only issue we encountered was the boxes not opening/running (you can tell if a servo is running by hearing it hum). Below are our solutions to this issue in order of likeliness:

1. The 9V battery needs to be replaced
2. One of the wires went loose, make sure everything is wire accurate to the provided schematic.
3. The servos for the boxes that are not opening are broken, and need to be replaced.

Maintenance:

To ensure the longevity and proper functioning of the tool, follow these maintenance guidelines:

(Before starting, carefully remove the shredded paper from the container and place it in another box or container to clear the workspace.)

- **Electrical components** _ Inspect the cardboard box:
 - Open the cardboard box to check the electrical components inside, including the Arduino microcontroller, wires, and breadboard.
 - Look for loose connections or damaged parts and fix or replace them if needed.
 - Close and tape the box carefully after the inspection.
- **Servo motors:** periodically check the connections to ensure that they remain securely attached and undamaged.
- **Keypad:** Test the buttons on the keypad to ensure they respond correctly, check for any signs of wear or damage. If a button is stuck or unresponsive, clean around it and test again. Replace the keypad if it continues to malfunction.
- **Battery:** If the 9V batteries are low or dead, replace them with new ones to keep the system running.
- **Plastic container:** can be replaced if there are cracks or damages.
- **Shredded paper:** refresh if it becomes worn out or too messy.

Note: Clean the keypad and external surfaces with a soft, dry cloth to remove any dirt or debris, and keep the prototype in a dry environment to prevent moisture damage to electronic components.

Feedback Description:

During the Quartz Hill Elementary school showcase, we got to present our final product to the people who attended the event. This gave us the opportunity to collect feedback from parents, students, and teachers in the form of an interview. All of the participants said that they thoroughly enjoyed playing our game and solving our puzzle. They believed that a sixth grader could benefit from this activity and have fun while learning about STEM related topics. They also added that the pyramid puzzle was challenging but made them think outside the box in order to solve it. When asked if they had any suggestions to improve our product, the majority stated that they would make the boxes close automatically after the puzzle piece was removed. They also said that they would reduce the size of the container as smaller students were not able to see inside. In the end, our product had high levels of enjoyment and interaction which met the goals we had set for our game.

Future Changes:

Although our product was a success, there are some aspects that could have been improved. To close the white boxes with the questions on top, you have to press the asterisk on the keypad which was repeatedly overlooked. In order to improve this function, we could add a sensor that closes our boxes once the puzzle piece is taken out. In the future we would also create a compartment for our wiring that is placed outside of the whole container. This would make it easier to maintain if anything is to need fixing and leave more space to place the boxes.

Overall Expenses:

Table 1. Bill of Materials

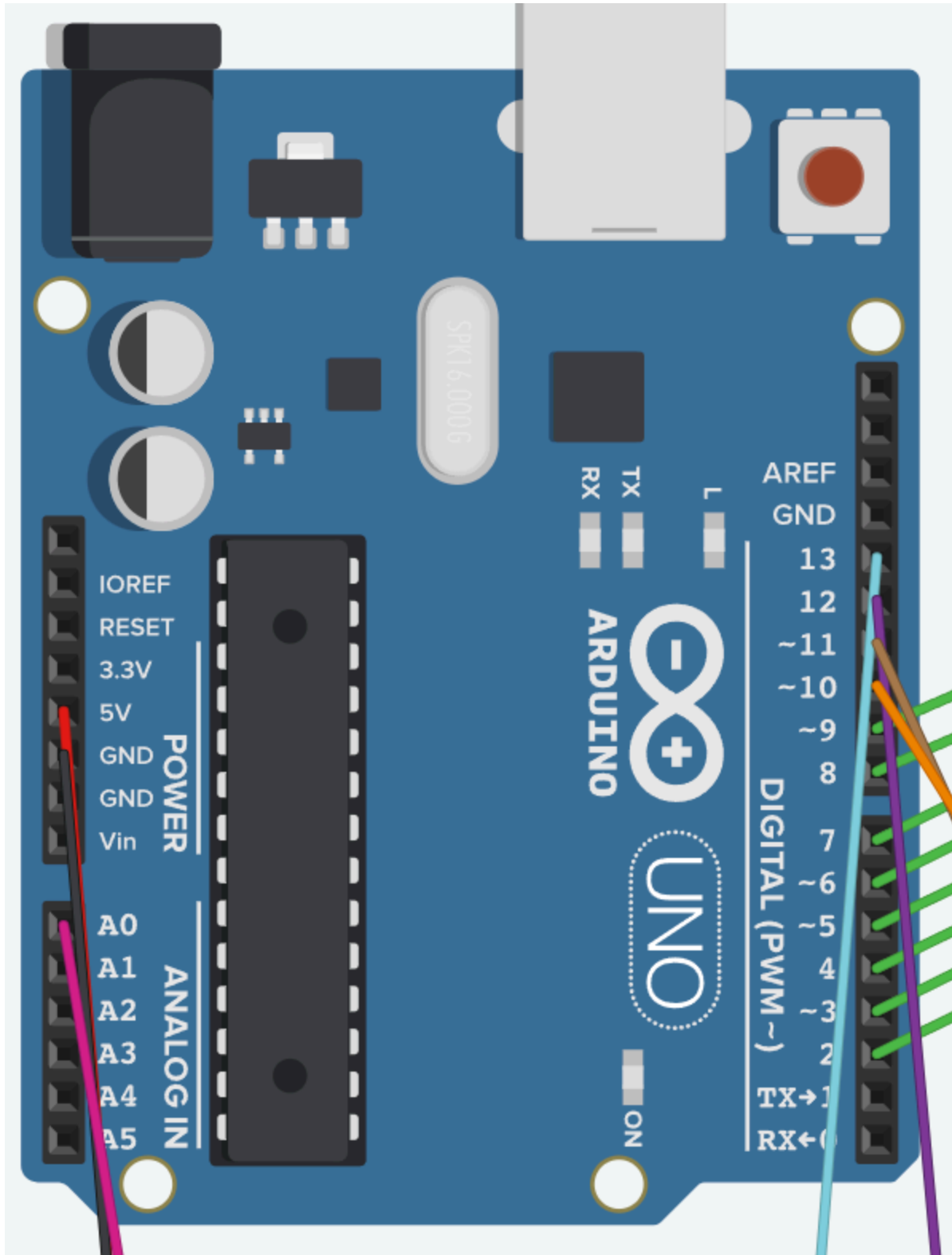
	Items	Price per unit (\$)	Quantities	Total Cost (\$)	Source
1	Plastic Bin	7.96 per	1	\$7.96	Amazon
2	Kinetic Sand	0.27 per ounce	1	\$14.22	Amazon
3	PLA filament	0.12/ per gram	Not sure yet	\$60.00	Innovation Hub
4	Servo motor	1.9 per count	10 (sell in pack)	\$18.99	Amazon
5	Arduino Microcontroller (Arduino Uno REV3)	27.60 per count	1	\$27.60	Amazon
6	Button(12x12x7.3 mm)	8.68	25 (sell in pack)	\$8.68	Amazon
7	Clear Plastic Basket Bags	1.25	1	\$1.25	Dollar Tree

8	Brown Shredded Paper	1.25	12	\$15.00	Dollar Tree
9	Amazon Basics 9V Alkaline Battery	2.26 per count	4 (sell in pack)	\$9.02	Amazon
10	Keypad 4x4	10.99	2 (sell in pack)	\$10.99	Amazon
11	Clear Storage Box	7	1	\$7.00	Target
Total:				\$180.71	

Our team successfully designed and constructed this interactive learning tool while staying well under the budget constraint of \$200. We kept the total project expenses at \$180.71 - saving approximately \$19.29 by smart shopping and using affordable materials from cost-effective retailers like Amazon, Dollar Tree, and the Innovation Hub.

Schematics:





Disclaimer: Colors of wires on schematics do not accurately reflect actual wires used.